



# CHASE S. FOSTER

UC Berkeley MIDS Graduate Student

📍 Athens, Tennessee 📞 423-300-3841 ✉️ chasefoster319@gmail.com

🌐 <https://chasefoster3.github.io/>  chasestephenfoster

## EDUCATION

2025 - Present  
University of  
California, Berkeley

### Master of Information and Data Science

- Completed Graduate Coursework: Experimentation and Causal Inference, Applied Machine Learning, Fundamentals of Data Engineering, Research Design, Data Science Programming
- In-Progress Coursework: Natural Language Processing with Deep Learning, Machine Learning Systems Engineering

2021 - 2025  
East Tennessee  
State University

### Bachelor of Science in Mathematics/Statistics

- Graduated Summa Cum Laude
- Major: Mathematics – Statistics Concentration
- Minors: Computer and Information Science and Global Citizenship Honors
- GPA: 3.976

## TECHNICAL SKILLS

**Programming:** Python, R, SQL, Java, C#

**Machine Learning and Statistics:** Regression, Classification, Clustering, Random Forest, Neural Networks, Dimensionality Reduction, Statistical Inference, Hypothesis Testing, Experimental Design, Power Analysis

**Data Science:** Exploratory Data Analysis, Data Visualization, Data Cleaning, Feature Engineering

**AI/NLP:** LLM Evaluation, Embeddings, Semantic Similarity, Sentence Transformers

**Libraries & Frameworks:** Scikit-learn, TensorFlow/Keras, Pandas, NumPy

**Tools:** Git, GitHub, Jupyter, RStudio, VSCode, Docker

## RELEVANT EXPERIENCE

2023  
Kansas City Royals  
Dominican Academy

### Summer Data Analytics Intern

- Conducted an independent research project analyzing Latin American players on MLB 40-man rosters.
- Collected and analyzed baseball information while developing familiarity with professional baseball data-collection technologies including Rapsodo and TrackMan.
- Collaborated with baseball operations and video staff in a professional sports environment.
- Supported game and practice operations while gaining experience with video collection, data-collection equipment, and day-to-day professional workflows.

2022 - 2025  
ETSU Center for  
Academic  
Achievement

### Mathematics & Computer Science Tutor

- Provided individual and group instruction in probability and statistics, calculus, and computer science to undergraduate students.
- Explained quantitative and computational concepts to students with varying levels of technical experience.
- Developed technical communication and analytical problem-solving skills through sustained instruction and mentorship.

## PROJECTS

---

### **MLB Postseason Prediction - ETSU - Honors Thesis**

- Analyzed 1998–2024 MLB team-season data to identify statistical factors associated with postseason success.
- Evaluated multiple statistical and machine learning approaches, including PCA, clustering, logistic regression, LDA, decision trees, bagging, and Random Forest.
- Identified ERA+, Runs Allowed, OPS+, and Runs Scored among the most important variables for distinguishing postseason outcomes.
- Selected a Random Forest model to generate predictions for the 2025 MLB postseason.

### **LLM Consistency Experiment - UC Berkeley MIDS - Experimentation & Causal Inference**

- Designed and conducted a controlled experiment investigating whether consistency-focused prompting affects the consistency of repeated LLM responses.
- Evaluated Gemini 2.5 Flash, GPT-4.1 Mini, and Gemma-3 across text and image-based tasks using repeated generations under controlled experimental conditions.
- Applied sentence-transformer embeddings (all-mpnet-base-v2) and cosine similarity to quantify semantic consistency across model responses.
- Analyzed treatment effects using regression models, interaction terms, and cluster-robust standard errors to evaluate differences across models and experimental conditions.
- Found that the consistency intervention reduced overall response consistency by approximately 20%, with effects varying substantially across models.

### **2026 FIFA World Cup Prediction - UC Berkeley MIDS - Applied Machine Learning**

- Integrated 49,071 international match records and 13,130 FIFA ranking records, producing a modeling dataset of 25,691 observations.
- Engineered 15 pre-match features to predict international soccer match outcomes while avoiding information unavailable before matches.
- Compared logistic regression, neural networks, and gradient boosting models to evaluate predictive performance.
- Achieved 60.89% test accuracy with logistic regression and used model predictions to simulate the 2026 FIFA World Cup.

### **NFL Officiating Exploratory Data Analysis - UC Berkeley MIDS - Data Science Programming**

- Conducted exploratory analysis of NFL penalties, referee assignments, replay reviews, and game-level data from 2015–2024.
- Integrated multiple datasets while cleaning inconsistent team, referee, date, and game identifiers.
- Investigated trends in penalty frequency, referee behavior, replay reviews, and officiating patterns across seasons and teams.
- Developed visualizations and statistical summaries to communicate trends and identify relationships within the data.

## LEADERSHIP

---

ETSU Army ROTC  
2021 - 2022

### **Contracted Cadet**

- Served as team and squad leader during tactical field-training exercises.
- Trained to lead teams and make decisions in simulated high-stress environments.
- Led preparation and execution of morning physical training activities.